

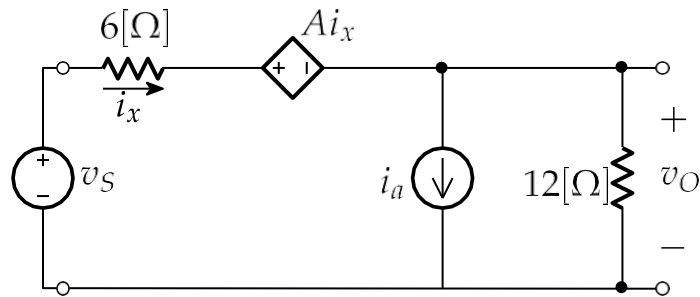
Homework 0

Problem 1

Use superposition to solve this problem. Choose values for i_a and A so that

$$v_o = 1/2 v_s - 3[\text{V}]$$

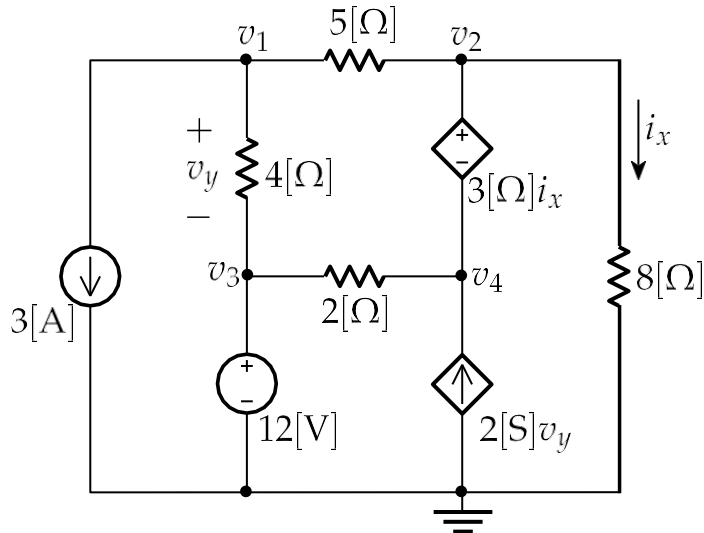
Hint: first find A by using only the voltage source, and then find the value of i_a by using only the current source.



Ans: $A = 6\Omega$; $i_a = 0.5\text{A}$

Problem 2

For the following circuit, use the node voltage method to determine the values of v_1 , v_2 , v_3 , and v_4 with respect to the reference node.

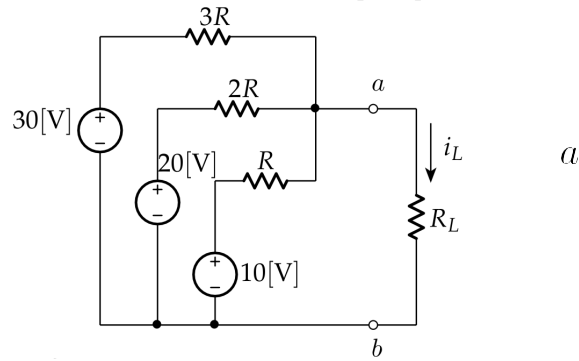


Ans: $v_1 = 23.5\text{V}$, $v_2 = 52.9\text{V}$, $v_3 = 12\text{V}$, and $v_4 = 33.1\text{V}$

Problem 3

For the following circuit, you measure an open circuit potential $v_{oc} = 180/11$ [V] between nodes a and b when you remove the load resistor. When you attach a load resistor $R_L = 40[\Omega]$, you

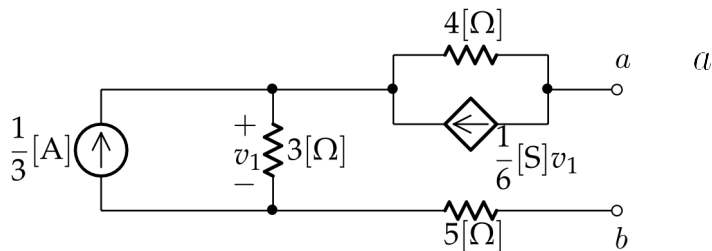
measure a current $i_L = 163.6[\text{mA}]$. Find R .



Ans: $110\ \Omega$

Problem 4

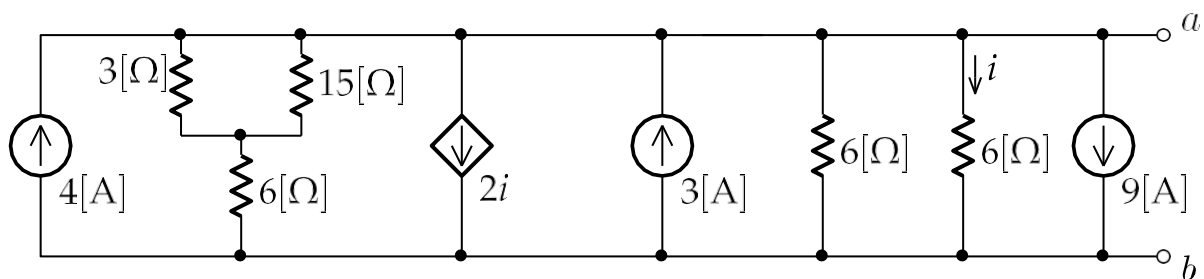
Find the Thevenin equivalent for the following circuit between nodes a and b .



Ans: $1/3\text{V}$; $10\ \Omega$

Problem 5

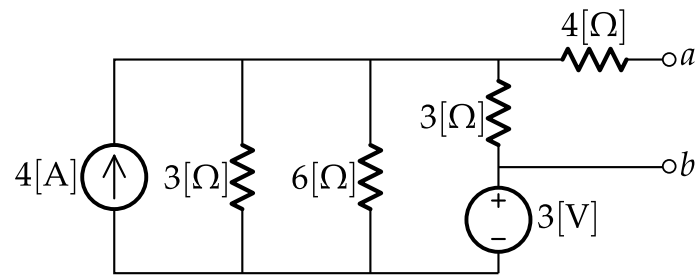
Find the Thevenin or Norton equivalent for the circuit below between nodes a and b .



Ans: -2.55V ; $1.275\ \Omega$; 2A

Problem 6

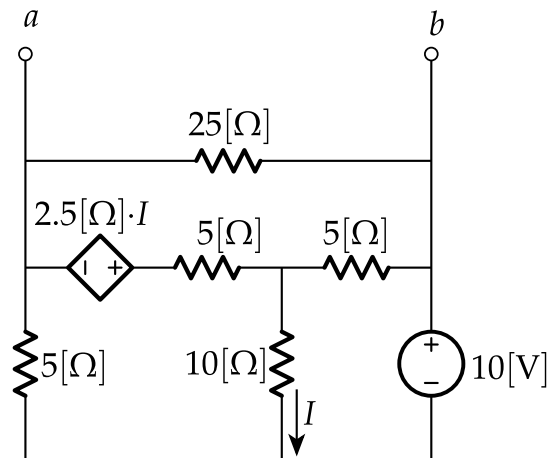
Find the Thévenin equivalent circuit at terminals (a,b) in the circuit below.



Ans: 3V; $26/5 \Omega$

Problem 7

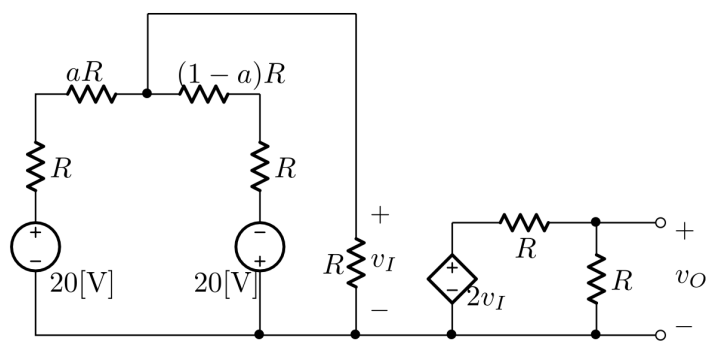
Find the Thévenin equivalent circuit at terminals (a,b) for the circuit below



Ans: $-50/7V$; $=75/28\Omega$

Problem 8

For the following circuit determine the value of v_o for $a=1/2$ and $a=1/4$

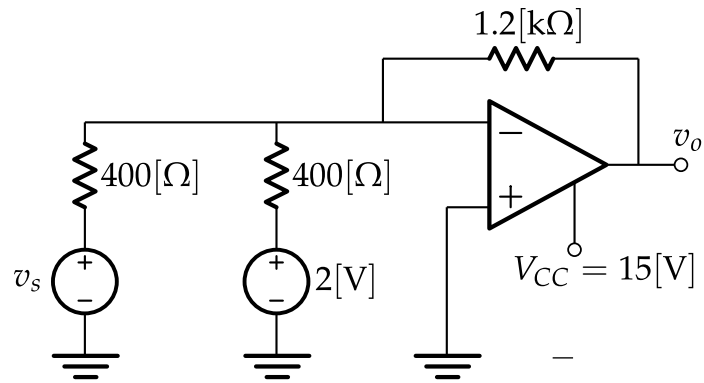


Ans: 0, 1.93V

Problem 9

- Determine the output voltage (v_o) in the circuit below in terms of v_s

- b. Specify the linear range for v_s



Ans: $-3v_s - 6$ [V] for -7 [V] $< v_s < 3$ [V]